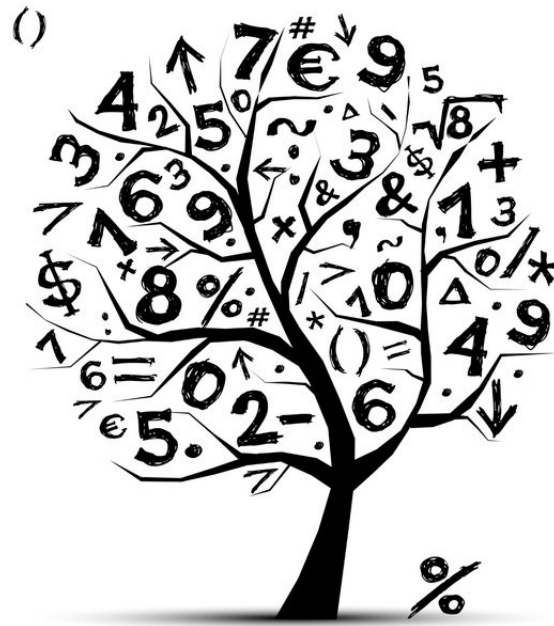


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System of linear equations



Contents

1	Existence and uniqueness of solutions	2
2	Gaussian Elimination Method	3
3	Gaussian elimination and the Rouché–Capelli theorem	6
4	Homogeneous systems	10
5	LU Factorization Method	12
6	Cramer’s rule	15
6.1	Systems of two equations and two variables	15
6.2	Systems of three equations and three variables	17
6.3	Computational cost	18

1 Existence and uniqueness of solutions

Consider a system of m linear equations with n variables:

$$A \mathbf{x} = \mathbf{b}$$

where:

- $A \in \mathbb{R}^{m \times n}$ is the coefficient matrix (m equations, n variables)
- $\mathbf{x} \in \mathbb{R}^{n \times 1}$ is the vector of variables
- $\mathbf{b} \in \mathbb{R}^{m \times 1}$ is the right-hand side vector
- $(A|\mathbf{b}) \in \mathbb{R}^{m \times (n+1)}$ is the augmented matrix

Rouché–Capelli theorem. The existence and uniqueness of solutions to the system $A \mathbf{x} = \mathbf{b}$ depend on the rank of the coefficient matrix A and the rank of the augmented matrix $(A|\mathbf{b})$.

Case 1 - No solution (inconsistent system):

$$\text{rank}(A) < \text{rank}(A|\mathbf{b})$$

The system is inconsistent and has no solution.

Case 2 - Unique solution:

$$\text{rank}(A) = \text{rank}(A|\mathbf{b}) = n$$

The system has a unique solution.

Case 3 - Infinite solutions:

$$\text{rank}(A) = \text{rank}(A|\mathbf{b}) < n$$

The system has infinitely many solutions with $n - \text{rank}(A)$ degrees of freedom (free variables). If $r = \text{rank}(A)$, the solutions can be described by means of $n - r$ **free parameters**.

- Since $(A|\mathbf{b})$ is obtained from A by adding one column, we always have

$$\text{rank}(A) \leq \text{rank}(A|\mathbf{b}) \leq \text{rank}(A) + 1,$$

so the three cases above cover all the possibilities. In particular, the system has at least one solution (it is **consistent**) if and only if $\text{rank}(A) = \text{rank}(A|\mathbf{b})$.

- Since $\text{rank}(A) \leq \min\{m, n\}$, a unique solution is possible only if $m \geq n$.

- If $A \in \mathbb{R}^{n \times n}$ is square, recall that $\text{rank}(A) = n$ if and only if $\det(A) \neq 0$. Hence, if $\det(A) \neq 0$, then also $\text{rank}(A|b) = n$ and the system has a unique solution, for every right-hand side b .

2 Gaussian Elimination Method

The **Gaussian elimination method** is an algorithm for solving systems of linear equations by transforming the augmented matrix $(A|b)$ into **row echelon form** (upper triangular form) through elementary row operations.

- Given the system of linear equations:

$$Ax = b$$

we form the **augmented matrix**:

$$(A|b)$$

- The method consists of two phases:

Phase 1 - Forward elimination: Transform the augmented matrix into upper triangular form using elementary row operations:

- Swap two rows
- Multiply a row by a non-zero scalar
- Add a multiple of one row to another row

The goal is to create zeros below the diagonal, obtaining:

$$\left(\begin{array}{cccc|c} \tilde{a}_{11} & \tilde{a}_{12} & \cdots & \tilde{a}_{1n} & \tilde{b}_1 \\ 0 & \tilde{a}_{22} & \cdots & \tilde{a}_{2n} & \tilde{b}_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \tilde{a}_{nn} & \tilde{b}_n \end{array} \right)$$

Phase 2 - Backward substitution: Solve the upper triangular system from bottom to top (assuming $\tilde{a}_{ii} \neq 0$ for all i):

$$\begin{aligned} x_n &= \frac{\tilde{b}_n}{\tilde{a}_{nn}} \\ x_{n-1} &= \frac{\tilde{b}_{n-1} - \tilde{a}_{n-1,n} x_n}{\tilde{a}_{n-1,n-1}} \\ &\vdots \\ x_i &= \frac{\tilde{b}_i - \sum_{j=i+1}^n \tilde{a}_{ij} x_j}{\tilde{a}_{ii}} \end{aligned}$$

Elementary row operations:

- $R_i \leftrightarrow R_j$: swap row i with row j
- $R_i \leftarrow k R_i$: multiply row i by scalar $k \neq 0$
- $R_i \leftarrow R_i + k R_j$: add k times row j to row i

These operations do not change the solution set of the system.

Example 1: Solving a system using Gaussian elimination - Part 1

Consider the system of linear equations:

$$\begin{cases} 2x_1 + 1x_2 + 1x_3 = 4 \\ 4x_1 + 3x_2 + 3x_3 = 10 \\ 8x_1 + 7x_2 + 9x_3 = 24 \end{cases}$$

Augmented matrix:

$$(A|\mathbf{b}) = \left(\begin{array}{ccc|c} 2 & 1 & 1 & 4 \\ 4 & 3 & 3 & 10 \\ 8 & 7 & 9 & 24 \end{array} \right)$$

Step 1: Eliminate x_1 from rows 2 and 3

$R_2 \leftarrow R_2 - 2R_1$ (subtract 2 times row 1 from row 2):

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 8 & 7 & 9 & 24 \end{array} \right)$$

$R_3 \leftarrow R_3 - 4R_1$ (subtract 4 times row 1 from row 3):

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 3 & 5 & 8 \end{array} \right)$$

Step 2: Eliminate x_2 from row 3

$R_3 \leftarrow R_3 - 3R_2$ (subtract 3 times row 2 from row 3):

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 2 & 2 \end{array} \right)$$

Upper triangular form achieved! The corresponding system is:

$$\begin{cases} 2x_1 + 1x_2 + 1x_3 = 4 \\ 1x_2 + 1x_3 = 2 \\ 2x_3 = 2 \end{cases}$$

Example 2: Solving a system using Gaussian elimination - Part 2

Backward substitution:

From the third equation:

$$2x_3 = 2 \implies x_3 = 1$$

From the second equation:

$$x_2 + x_3 = 2 \implies x_2 = 2 - x_3 = 2 - 1 = 1$$

From the first equation:

$$2x_1 + x_2 + x_3 = 4 \implies 2x_1 = 4 - x_2 - x_3 = 4 - 1 - 1 = 2 \implies x_1 = 1$$

Solution:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Verification:

$$\mathbf{A}\mathbf{x} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2+1+1 \\ 4+3+3 \\ 8+7+9 \end{pmatrix} = \begin{pmatrix} 4 \\ 10 \\ 24 \end{pmatrix} = \mathbf{b} \quad \checkmark$$

3 Gaussian elimination and the Rouché–Capelli theorem

- Gaussian elimination can be applied to any system $A \mathbf{x} = \mathbf{b}$, with $A \in \mathbb{R}^{m \times n}$. In general, forward elimination transforms $(A|\mathbf{b})$ into **row echelon form**: in each nonzero row, the first nonzero entry (called **pivot**) is strictly to the right of the pivot of the row above, and the rows of zeros (if any) are at the bottom.
- Elementary row operations do not change the solution set of the system, and they do not change the rank of A nor the rank of $(A|\mathbf{b})$.
- The rank of a matrix in row echelon form is equal to the number of its nonzero rows. Therefore, at the end of forward elimination we can read $\text{rank}(A)$ and $\text{rank}(A|\mathbf{b})$ and apply the Rouché–Capelli theorem:
 - a row of the form $(0 \ \cdots \ 0 \mid c)$ with $c \neq 0$ corresponds to the impossible equation $0 = c$: then $\text{rank}(A) < \text{rank}(A|\mathbf{b})$ and the system has no solution;
 - otherwise, the variables whose columns contain a pivot are computed by backward substitution, and the remaining $n - r$ variables are the **free parameters**.

When the system has infinitely many solutions with one free parameter $t \in \mathbb{R}$, the solutions can be written in **parametric form**

$$\mathbf{x} = \mathbf{x}_0 + t \mathbf{v}, \quad t \in \mathbb{R},$$

where $\mathbf{x}_0$ is a particular solution (obtained for $t = 0$) and $\mathbf{v}$ is a nonzero vector such that $A \mathbf{v} = \mathbf{0}$.

Example 3: Rouché–Capelli: unique solution

Consider the system of linear equations:

$$\begin{cases} 1 x_2 + 1 x_3 = 1 \\ 1 x_1 + 1 x_2 + 1 x_3 = 2 \\ 2 x_1 + 1 x_2 + 3 x_3 = 1 \end{cases} \quad (A|\mathbf{b}) = \left(\begin{array}{ccc|c} 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 2 & 1 & 3 & 1 \end{array} \right)$$

The entry in position (1, 1) is zero, so we first swap rows 1 and 2.

$R_1 \leftrightarrow R_2$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 2 & 1 & 3 & 1 \end{array} \right)$$

$$R_3 \leftarrow R_3 - 2 R_1:$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & -1 & 1 & -3 \end{array} \right)$$

$$R_3 \leftarrow R_3 + R_2:$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & -2 \end{array} \right)$$

There are 3 nonzero rows both in A and in $(A|\mathbf{b})$: $\text{rank}(A) = \text{rank}(A|\mathbf{b}) = 3 = n$, hence the solution is unique. By backward substitution:

$$2 x_3 = -2 \implies x_3 = -1, \quad x_2 = 1 - x_3 = 2, \quad x_1 = 2 - x_2 - x_3 = 2 - 2 + 1 = 1,$$

that is,

$$\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}.$$

Example 4: Rouché–Capelli: infinitely many solutions

Consider the system of linear equations:

$$\begin{cases} 1x_1 + 2x_2 - 1x_3 = 1 \\ 2x_1 + 5x_2 + 1x_3 = 4 \\ 3x_1 + 7x_2 = 5 \end{cases} \quad (A|b) = \left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & 5 & 1 & 4 \\ 3 & 7 & 0 & 5 \end{array} \right)$$

$R_2 \leftarrow R_2 - 2R_1$:

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 3 & 7 & 0 & 5 \end{array} \right)$$

$R_3 \leftarrow R_3 - 3R_1$:

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 1 & 3 & 2 \end{array} \right)$$

$R_3 \leftarrow R_3 - R_2$:

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

There are 2 nonzero rows both in A and in $(A|b)$: $\text{rank}(A) = \text{rank}(A|b) = 2 < 3 = n$, hence there are infinitely many solutions with $n - r = 3 - 2 = 1$ free parameter. The pivots are in the columns of x_1 and x_2 , so x_3 is free: we set $x_3 = t$, $t \in \mathbb{R}$. By backward substitution:

$$x_2 = 2 - 3x_3 = 2 - 3t, \quad x_1 = 1 - 2x_2 + x_3 = 1 - 4 + 6t + t = -3 + 7t.$$

In parametric form:

$$\mathbf{x} = \begin{pmatrix} -3+7t \\ 2-3t \\ t \end{pmatrix} = \underbrace{\begin{pmatrix} -3 \\ 2 \\ 0 \end{pmatrix}}_{\mathbf{x}_0} + t \underbrace{\begin{pmatrix} 7 \\ -3 \\ 1 \end{pmatrix}}_{\mathbf{v}}, \quad t \in \mathbb{R}.$$

Verification: $A\mathbf{x}_0 = (-3+4, -6+10, -9+14)' = (1, 4, 5)' = \mathbf{b}$ and $A\mathbf{v} = (7-6-1, 14-15+1, 21-21)' = \mathbf{0}$, hence $A(\mathbf{x}_0 + t\mathbf{v}) = \mathbf{b}$ for every $t \in \mathbb{R}$ ✓

Example 5: Rouché–Capelli: no solution

We change only the last right-hand side of the previous system:

$$\begin{cases} 1x_1 + 2x_2 - 1x_3 = 1 \\ 2x_1 + 5x_2 + 1x_3 = 4 \\ 3x_1 + 7x_2 = 6 \end{cases} \quad (A|b) = \left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & 5 & 1 & 4 \\ 3 & 7 & 0 & 6 \end{array} \right)$$



With the same row operations $R_2 \leftarrow R_2 - 2 R_1$, $R_3 \leftarrow R_3 - 3 R_1$:

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 1 & 3 & 3 \end{array} \right)$$

$R_3 \leftarrow R_3 - R_2$:

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 1 \end{array} \right)$$

The last row corresponds to the equation $0 x_1 + 0 x_2 + 0 x_3 = 1$, which is impossible. Indeed A has 2 nonzero rows while $(A|\mathbf{b})$ has 3:

$$\text{rank}(A) = 2 < 3 = \text{rank}(A|\mathbf{b}),$$

hence the system is inconsistent and has **no solution**.

4 Homogeneous systems

Definition 1: homogeneous system

A system of linear equations $A \mathbf{x} = \mathbf{b}$ with $A \in \mathbb{R}^{m \times n}$ is called **homogeneous** if $\mathbf{b} = \mathbf{0}$, i.e., if it has the form

$$A \mathbf{x} = \mathbf{0}.$$

Observation 1: solutions of a homogeneous system

Let $A \in \mathbb{R}^{m \times n}$. Then:

- the homogeneous system $A \mathbf{x} = \mathbf{0}$ always has the solution $\mathbf{x} = \mathbf{0}$, called the **trivial solution**;
- it has nontrivial solutions ($\mathbf{x} \neq \mathbf{0}$) if and only if $\text{rank}(A) < n$; in this case it has infinitely many solutions, with $n - \text{rank}(A)$ free parameters;
- if $A \in \mathbb{R}^{n \times n}$ is square, it has nontrivial solutions if and only if $\det(A) = 0$.

Proof. Adding a column of zeros does not change the rank, hence $\text{rank}(A|\mathbf{0}) = \text{rank}(A)$ and the system is always consistent (as shown by $\mathbf{x} = \mathbf{0}$). By the Rouché–Capelli theorem, the solution $\mathbf{x} = \mathbf{0}$ is the unique one if and only if $\text{rank}(A) = n$; otherwise there are infinitely many solutions with $n - \text{rank}(A)$ free parameters. For a square matrix, $\text{rank}(A) < n$ if and only if $\det(A) = 0$. $\square$

- In particular, if $m < n$ (fewer equations than variables), then $\text{rank}(A) \leq m < n$ and the homogeneous system always has nontrivial solutions.
- If $\mathbf{x}_0$ is a solution of $A \mathbf{x} = \mathbf{b}$ and $\mathbf{v}$ is a solution of $A \mathbf{x} = \mathbf{0}$, then $A(\mathbf{x}_0 + \mathbf{v}) = \mathbf{b} + \mathbf{0} = \mathbf{b}$. This explains the parametric form $\mathbf{x} = \mathbf{x}_0 + t \mathbf{v}$ of Example 4.

Example 6: a homogeneous system

Consider the homogeneous system with the coefficient matrix of Example 4:

$$\begin{cases} 1x_1 + 2x_2 - 1x_3 = 0 \\ 2x_1 + 5x_2 + 1x_3 = 0 \\ 3x_1 + 7x_2 = 0 \end{cases}$$

By the Sarrus rule,

$$\det(A) = 1 \cdot 5 \cdot 0 + 2 \cdot 1 \cdot 3 + (-1) \cdot 2 \cdot 7 - (-1) \cdot 5 \cdot 3 - 2 \cdot 2 \cdot 0 - 1 \cdot 1 \cdot 7 = 0 + 6 - 14 + 15 - 0 - 7 = 0,$$

so there are nontrivial solutions. With the same row operations as in Example 4 (the last column stays

equal to zero):

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Setting $x_3 = t$: $x_2 = -3t$ and $x_1 = -2x_2 + x_3 = 6t + t = 7t$. Hence the solutions are

$$\mathbf{x} = t \begin{pmatrix} 7 \\ -3 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R},$$

and the trivial solution is obtained for $t = 0$.

5 LU Factorization Method

The **LU factorization** (or **LU decomposition**) is a method for solving systems of linear equations by decomposing the coefficient matrix A into the product of two triangular matrices:

$$A = LU$$

where L is a **lower triangular matrix** (with 1's on the diagonal) and U is an **upper triangular matrix**.

- Given the system of linear equations:

$$Ax = b$$

if we have the LU factorization $A = LU$, we can substitute:

$$LUx = b$$

- We introduce the **auxiliary variable** y defined as:

$$y := Ux$$

so that $LUx = b$ becomes $Ly = b$.

Why does this work? We are splitting the original system $Ax = b$ into two simpler triangular systems:

- Find y such that $Ly = b$.
- Find x such that $Ux = y$.

If both steps succeed, then $Ax = LUx = Ly = b$, so x is indeed a solution of the original system. Since both L and U are triangular, each of the two systems can be solved in $O(n^2)$ operations by simple substitution.

- We then solve the system in two steps:

Step 1 - Forward substitution: Solve $Ly = b$ for y

Since L is lower triangular, we can solve this system easily from top to bottom:

$$\begin{aligned} y_1 &= b_1 \\ y_2 &= b_2 - \ell_{21} y_1 \\ y_3 &= b_3 - \ell_{31} y_1 - \ell_{32} y_2 \\ &\vdots \\ y_i &= b_i - \sum_{j=1}^{i-1} \ell_{ij} y_j \end{aligned}$$

Step 2 - Backward substitution: Solve $\mathbf{U} \mathbf{x} = \mathbf{y}$ for $\mathbf{x}$

Since $\mathbf{U}$ is upper triangular, we can solve this system easily from bottom to top:

$$\begin{aligned} x_n &= \frac{y_n}{u_{nn}} \\ x_{n-1} &= \frac{y_{n-1} - u_{n-1,n} x_n}{u_{n-1,n-1}} \\ &\vdots \\ x_i &= \frac{y_i - \sum_{j=i+1}^n u_{ij} x_j}{u_{ii}} \end{aligned}$$

Advantages of LU factorization:

- Once the factorization $\mathbf{A} = \mathbf{L} \mathbf{U}$ is computed, we can solve the system for different right-hand sides $\mathbf{b}$ efficiently
- Both forward and backward substitution require only $O(n^2)$ operations
- The factorization itself requires $O(n^3)$ operations, but it needs to be done only once

Example 7: Solving a system using LU factorization - Part 1

Consider the system of linear equations:

$$\begin{cases} 2x_1 + 1x_2 + 1x_3 = 4 \\ 4x_1 + 3x_2 + 3x_3 = 10 \\ 8x_1 + 7x_2 + 9x_3 = 24 \end{cases}$$

In matrix form: $\mathbf{A} \mathbf{x} = \mathbf{b}$ where

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 4 \\ 10 \\ 24 \end{pmatrix}$$

Given the LU factorization of $\mathbf{A}$:

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{pmatrix} \quad \mathbf{U} = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

Verification:

$$\mathbf{L} \mathbf{U} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix} = \mathbf{A} \quad \checkmark$$

Example 8: Solving a system using LU factorization - Part 2

Step 1 - Forward substitution: Solve $L y = b$

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 10 \\ 24 \end{pmatrix}$$

From the first equation:

$$y_1 = 4$$

From the second equation:

$$2 y_1 + y_2 = 10 \implies y_2 = 10 - 2 \cdot 4 = 10 - 8 = 2$$

From the third equation:

$$4 y_1 + 3 y_2 + y_3 = 24 \implies y_3 = 24 - 4 \cdot 4 - 3 \cdot 2 = 24 - 16 - 6 = 2$$

Thus:

$$\mathbf{y} = \begin{pmatrix} 4 \\ 2 \\ 2 \end{pmatrix}$$

Step 2 - Backward substitution: Solve $U x = y$

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 2 \end{pmatrix}$$

From the third equation:

$$2 x_3 = 2 \implies x_3 = 1$$

From the second equation:

$$x_2 + x_3 = 2 \implies x_2 = 2 - 1 = 1$$

From the first equation:

$$2 x_1 + x_2 + x_3 = 4 \implies 2 x_1 = 4 - 1 - 1 = 2 \implies x_1 = 1$$

Solution:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Verification:

$$\mathbf{A} \mathbf{x} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2+1+1 \\ 4+3+3 \\ 8+7+9 \end{pmatrix} = \begin{pmatrix} 4 \\ 10 \\ 24 \end{pmatrix} = \mathbf{b} \quad \checkmark$$

6 Cramer's rule

Cramer's rule is an explicit formula for the solution of a system of m linear equations with m variables (valid whenever the system has a unique solution).

- Given a column vector $\mathbf{b} \in \mathbb{R}^{m \times 1}$ of m rows, a matrix $\mathbf{A} \in \mathbb{R}^{m \times m}$ of m rows and m columns and a column vector $\mathbf{x} \in \mathbb{R}^{m \times 1}$ of m rows containing the m variables:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & \dots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mm} \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix}$$

if $\det(\mathbf{A}) \neq 0$, the solution $\tilde{\mathbf{x}}$ of the system of m linear equations:

$$\mathbf{A} \mathbf{x} = \mathbf{b}$$

is given by the formula:

$$\tilde{x}_j = \frac{\det(\mathbf{A}_j)}{\det(\mathbf{A})}, \quad \forall j \in \{1, 2, \dots, m\} \quad (1)$$

where $\mathbf{A}_j$ is the matrix formed by replacing the j -th column of $\mathbf{A}$ by the column vector $\mathbf{b}$.

6.1 Systems of two equations and two variables

With $m = 2$, we have:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \quad \mathbf{A}_1 = \begin{pmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{pmatrix} \quad \mathbf{A}_2 = \begin{pmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{pmatrix}$$

and

$$\det(\mathbf{A}) = a_{11} a_{22} - a_{12} a_{21}$$

$$\det(\mathbf{A}_1) = b_1 a_{22} - a_{12} b_2$$

$$\det(\mathbf{A}_2) = a_{11} b_2 - b_1 a_{21}$$

If $\det(\mathbf{A}) \neq 0$, we then have:

$$\begin{cases} a_{11} x_1 + a_{12} x_2 = b_1 \\ a_{21} x_1 + a_{22} x_2 = b_2 \end{cases} \implies (\tilde{x}_1, \tilde{x}_2) = \left(\frac{\det(\mathbf{A}_1)}{\det(\mathbf{A})}, \frac{\det(\mathbf{A}_2)}{\det(\mathbf{A})} \right)$$

Example 9: solution of a system of two equations and two variables

- Given

$$\mathbf{A} = \begin{pmatrix} -1 & 1 \\ 8 & 2 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 2 \\ 19 \end{pmatrix}$$

we have

$$\mathbf{A}_1 = \begin{pmatrix} 2 & 1 \\ 19 & 2 \end{pmatrix} \quad \mathbf{A}_2 = \begin{pmatrix} -1 & 2 \\ 8 & 19 \end{pmatrix}$$

and

$$\det(\mathbf{A}) = (-1) \cdot 2 - 1 \cdot 8 = -10$$

$$\det(\mathbf{A}_1) = 2 \cdot 2 - 1 \cdot 19 = -15$$

$$\det(\mathbf{A}_2) = (-1) \cdot 19 - 2 \cdot 8 = -35$$

Since $\det(\mathbf{A}) \neq 0$, we then have:

$$\begin{cases} -x_1 + x_2 = 2 \\ 8x_1 + 2x_2 = 19 \end{cases} \implies (\tilde{x}_1, \tilde{x}_2) = \left(\frac{-15}{-10}, \frac{-35}{-10} \right) = \left(\frac{3}{2}, \frac{7}{2} \right)$$

- Given

$$\mathbf{A} = \begin{pmatrix} -1 & -1 \\ 1 & -1 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}$$

we have

$$\mathbf{A}_1 = \begin{pmatrix} -2 & -1 \\ 0 & -1 \end{pmatrix} \quad \mathbf{A}_2 = \begin{pmatrix} -1 & -2 \\ 1 & 0 \end{pmatrix}$$

and

$$\det(\mathbf{A}) = (-1) \cdot (-1) - (-1) \cdot 1 = 2$$

$$\det(\mathbf{A}_1) = (-2) \cdot (-1) - (-1) \cdot 0 = 2$$

$$\det(\mathbf{A}_2) = (-1) \cdot 0 - (-2) \cdot 1 = 2$$

Since $\det(\mathbf{A}) \neq 0$, we then have:

$$\begin{cases} -x_1 - x_2 = -2 \\ x_1 - x_2 = 0 \end{cases} \implies (\tilde{x}_1, \tilde{x}_2) = \left(\frac{2}{2}, \frac{2}{2} \right) = (1, 1)$$

6.2 Systems of three equations and three variables

With $m = 3$, we have:

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

$$A_1 = \begin{pmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{pmatrix} \quad A_2 = \begin{pmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{pmatrix} \quad A_3 = \begin{pmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{pmatrix}$$

and (Sarrus rule)

$$\det(A) = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32}$$

$$\det(A_1) = b_1a_{22}a_{33} + a_{12}a_{23}b_3 + a_{13}b_2a_{32} - a_{13}a_{22}b_3 - a_{12}b_2a_{33} - b_1a_{23}a_{32}$$

$$\det(A_2) = a_{11}b_2a_{33} + b_1a_{23}a_{31} + a_{13}a_{21}b_3 - a_{13}b_2a_{31} - b_1a_{21}a_{33} - a_{11}a_{23}b_3$$

$$\det(A_3) = a_{11}a_{22}b_3 + a_{12}b_2a_{31} + b_1a_{21}a_{32} - b_1a_{22}a_{31} - a_{12}a_{21}b_3 - a_{11}b_2a_{32}$$

If $\det(A) \neq 0$, we then have:

$$(\tilde{x}_1, \tilde{x}_2, \tilde{x}_3) = \left(\frac{\det(A_1)}{\det(A)}, \frac{\det(A_2)}{\det(A)}, \frac{\det(A_3)}{\det(A)} \right)$$

Example 10: solution of a system of three equations and three variables

We solve with Cramer's rule the system of Example 1:

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 4 \\ 10 \\ 24 \end{pmatrix}$$

We have

$$A_1 = \begin{pmatrix} 4 & 1 & 1 \\ 10 & 3 & 3 \\ 24 & 7 & 9 \end{pmatrix} \quad A_2 = \begin{pmatrix} 2 & 4 & 1 \\ 4 & 10 & 3 \\ 8 & 24 & 9 \end{pmatrix} \quad A_3 = \begin{pmatrix} 2 & 1 & 4 \\ 4 & 3 & 10 \\ 8 & 7 & 24 \end{pmatrix}$$

and

$$\det(A) = 54 + 24 + 28 - 24 - 36 - 42 = 4$$

$$\det(A_1) = 108 + 72 + 70 - 72 - 90 - 84 = 4$$

$$\det(A_2) = 180 + 96 + 96 - 80 - 144 - 144 = 4$$

$$\det(A_3) = 144 + 80 + 112 - 96 - 96 - 140 = 4$$

Since $\det(\mathbf{A}) \neq 0$, we then have:

$$(\tilde{x}_1, \tilde{x}_2, \tilde{x}_3) = \left(\frac{4}{4}, \frac{4}{4}, \frac{4}{4} \right) = (1, 1, 1),$$

which is the solution found with Gaussian elimination.

6.3 Computational cost

- Cramer's rule gives an explicit formula, which is very useful for small systems ($m = 2$ or $m = 3$) and for theoretical purposes. However, it is **not** a practical method for large systems.
- To apply Cramer's rule we need $m + 1$ determinants of order m : $\det(\mathbf{A})$, $\det(\mathbf{A}_1)$, $\dots$, $\det(\mathbf{A}_m)$. If each determinant is computed with the Laplace expansion, the number of arithmetic operations grows roughly like $m!$ for each determinant.
- Gaussian elimination, instead, solves the system with a number of operations which grows roughly like m^3 (about $\frac{2}{3}m^3$ operations).
- For example, with $m = 10$ we have $10! = 3\,628\,800$, while $\frac{2}{3} \cdot 10^3 \approx 667$; with $m = 20$, $20!$ is larger than $2 \cdot 10^{18}$, while $\frac{2}{3} \cdot 20^3 \approx 5333$.
- Even if the determinants are computed in a more efficient way (for instance with Gaussian elimination itself), Cramer's rule still needs $m + 1$ of them, and it remains more expensive than solving the system directly with Gaussian elimination.